

1. PixelProof AI image detection

Reducing false AI warnings on real photographs

Efe Han Keleş
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Supervisor: Önder Çelebi, Director
Planning and Development Directorate

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PixelProof research prototype

2. The problem and objective

A real photograph can receive a false AI warning.

An AI image can also escape detection.

The objective was to reduce the first error without creating new misses.

3. Early accuracy did not transfer

Table 1 shows the first major warning.

Table 1. Historical ResNet-18 accuracy.

Image collection	Accuracy
Familiar test images	97.66%
Separate external collection	25.2%

The gap led to checks of resolution, sources and labels.

4. Transfer learning in PixelProof

Existing models describe the image

Frozen DINOv2, CLIP and DEAR components

Our trained components use those descriptions

E92 adjusts the detection score

Python and PyTorch for modelling
FastAPI and React for the local demo

5. Training and development data

Table 2 separates learning from repeated comparison.

Table 2. The main E92 data groups.

Purpose	Images	Use
Training	12,269	Learn the model
Development	160 real + 160 AI	Compare candidates

Four training views per image: 49,076 views
Repeated views do not create independent photographs.

6. Fewer false warnings on real images

Figure 1 compares the same 160 real images in each condition.

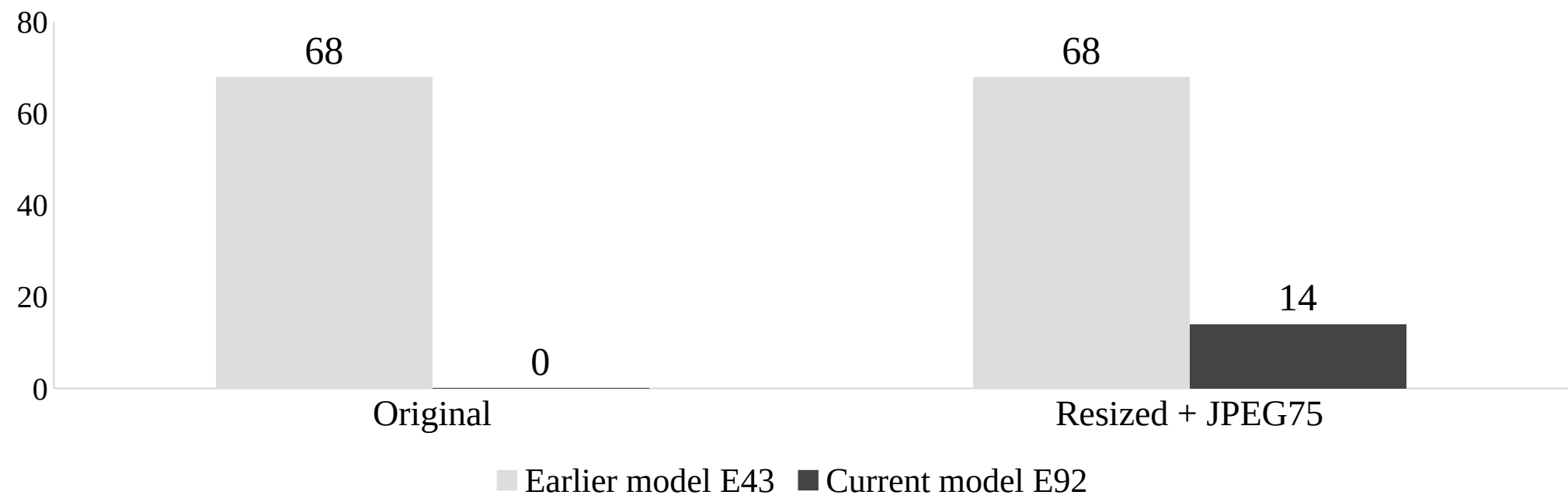


Figure 1. False AI warnings on reused development images.

E92 detected 159/160 AI images in each condition.

7. Full acceptance remained incomplete

20/20 numerical development checks passed

One AI image detected by E43 became a new miss.

That separate protection rule failed.

A fresh, independent final test is still needed.

8. The local web demonstration

An upload receives one of the outcomes in Table 3.

Table 3. How the demo communicates evidence.

Result	Meaning
AI evidence	The detector found a warning signal
No clear AI evidence	The detector found insufficient evidence
Uncertain	The checks do not support a clear outcome

The score measures detector response.
It does not certify the photograph’s authenticity.

9. Model2 remains experimental

Task: locate an AI-edited region

Data: source overlap and unclear edit masks

Local generation also changed background pixels.
Controlled pilot: 16 reused images, one editor.

The candidate was rejected.

10. Outcome and next step

Table 4 summarizes the outcome of the internship.

Table 4. Delivered work and remaining evidence.

Delivered	Next step
Working local prototype	Independent final evaluation
Fewer development false warnings	More cameras and generators
Recorded experiments and corrections	Broader Model2 controls

My main lesson: check what a good score actually proves.

11. References for the method

Radford et al. (2021)

Learning transferable visual models from natural language supervision. PMLR 139.

Oquab et al. (2024)

DINOv2: Learning robust visual features without supervision. TMLR.

Kim et al. (2026)

Dissect and prune: Enhancing robustness in AI-generated image detection. ICML.

Ojha et al. (2023)

Towards universal fake image detectors that generalize across generative models. CVPR.

12. References for data and evaluation

Abdelhamed et al. (2018)

A high-quality denoising dataset for smartphone cameras. CVPR.

Guo et al. (2017)

On calibration of modern neural networks. PMLR 70.

Dwork et al. (2015)

Generalization in adaptive data analysis and holdout reuse. NeurIPS 28.

Keleş (2026)

PixelProof experiment, history and dataset records. GitHub, commit 497d45c.